

ELECTIVE COURSE-6.3

URA413: FINITE ELEMENT METHODS				
	L	T	P	Cr
	3	0	0	3.0
Course Objective: To expose the students to the basics of Finite Element Methods				
Syllabus Introduction: Finite element methods, history and range of applications. Finite Elements: Definition and properties, assembly rules and general assembly procedure, features of assembled matrix, boundary conditions. Continuum Problems: Classification of differential equations, variational formulation approach, Ritz method, generalized definition of an element, element equations from variations. Galerkin's weighted residual approach, energy balance methods. Element Shapes and Interpolation Functions: Basic element shapes, generalized co-ordinates, polynomials, natural co-ordinates in one-, two- and three-dimensions, Lagrange and Hermite polynomials, two-D and three-D elements for C^0 and C^1 problems, co-ordinate transformation, iso-parametric elements and numerical integration, introduction to p and h type of formulations. Application of Finite Element Methods to elasticity problems and heat conduction Problems. Minor Project: Students will be given different components related to machines/structures and will be asked to analyze these components using ANSYS or related analysis software packages. Students will also be asked to make their own codes for simple problems using MATLAB and compare their results with any of the commercial packages. The components will be analyzed using different linear / higher order elements i.e., triangular, axisymmetric, quadrilateral, tetrahedral and hexahedral elements.				
Course Learning Outcomes (CLOs) The students will be able to: 1. apply the procedure involved to solve a problem using Finite Element Methods. 2. develop the element stiffness matrices using different approach. 3. analyze a 2D and 3D problem using different types of elements. 4. solve problems using the available commercial package.				
Text Books Chandrupatla, T. R. and Belegundu, A. K., Introduction to Finite Elements in Engineering, Pearson Education, India (2001). 1 Huebner, K. H., The Finite Element Method for Engineers, John Wiley, New York (2001).				
Reference Books 1 Bathe, K.J., Finite Element Procedure in Engineering Analysis, Englewood Cliffs, Prentice Hall, New York (2001). 2 Zienkiewicz, O. C., The Finite Element Methods, Tata McGraw Hill, New Delhi (2002).				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

- 3 Reddy, J. N., An Introduction to Finite Elements Methods, McGraw Hill, New York (2001).
- 4 Stasa, F.L., Applied Finite Element Analysis for Engineers, Holt, Rinehart and Winston, New York (1995).

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1	MST	30
2	EST	45
3	Sessionals (Minor Project/Tutorials/Quizzes)	25

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